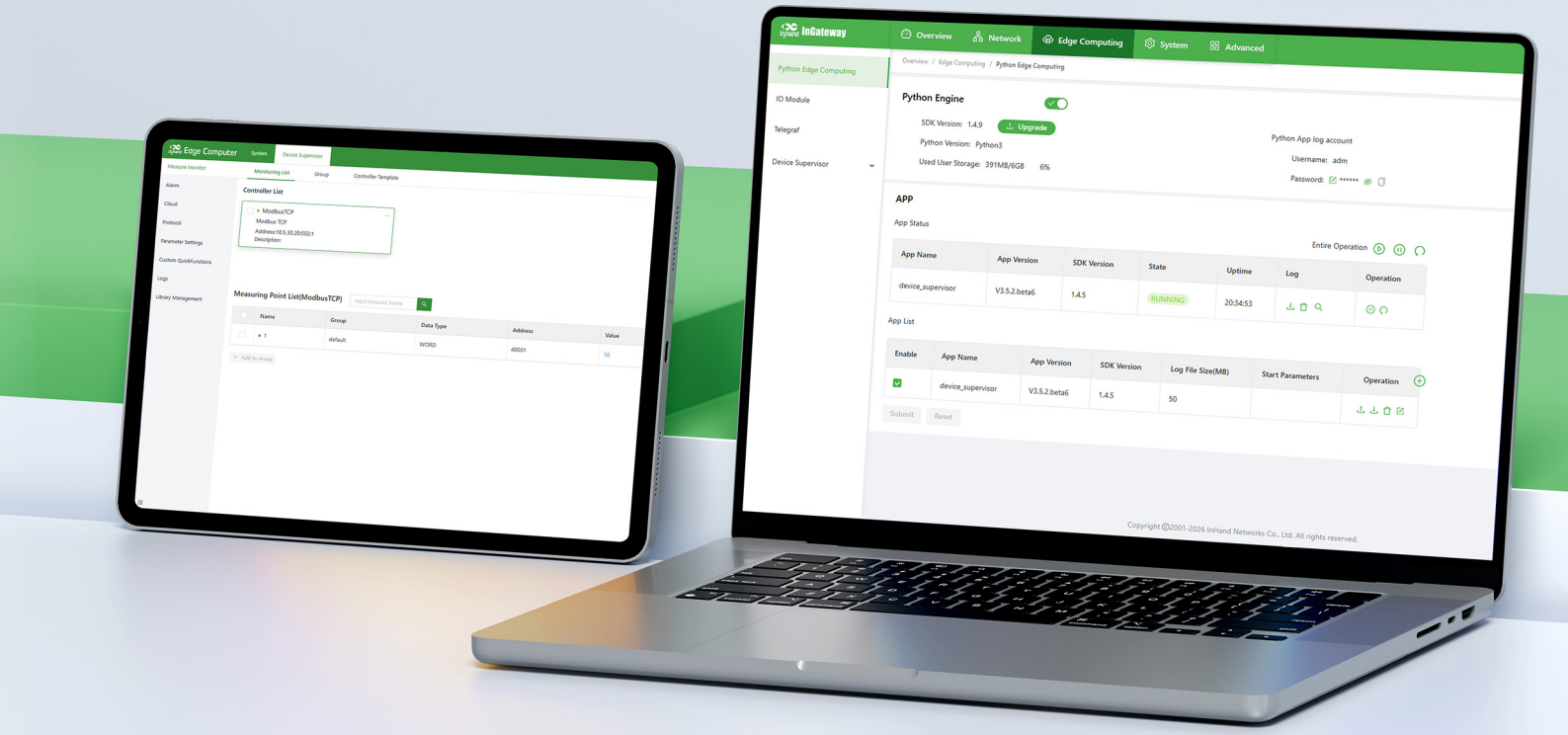
InHand Networks

Software-Defined Acquisition, Intelligence-Driven Edge



DeviceSupervisor™ Agent
Industrial Edge Agent Software

- 80+ Acquisition Protocols
- Multi-Cloud Integration
- Python Edge Intelligence
- Zero-Code / Low-Code

DeviceSupervisor™ Agent (DSA)

Industrial Edge Agent Software · Product Datasheet

Version: V1.0 **Based on Software Version**: DSA V3.1.10 **Release Date**: September 2026 **Vendor**: InHand Networks

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1. Product Overview

DeviceSupervisor™ Agent (DSA) is an **industrial edge agent software** developed by InHand Networks, purpose-built for InHand IG series industrial gateways and EC series edge computers. Through a **"zero-code / low-code"** approach, DSA empowers customers to rapidly deploy the following capabilities at the industrial edge:

- **Southbound Data Acquisition** — one-stop connectivity to nearly 100 industrial device protocols
- **Northbound Protocol Conversion** — expose acquired data via standard industrial protocols to upstream systems
- **Cloud Platform Integration** — real-time streaming to mainstream Industrial IoT cloud platforms
- **Edge Intelligence** — Python-scripted data filtering, aggregation, alarming, and analytics at the edge

DSA delivers **low-latency data access management** and **intelligent analytics services**, enabling customers to quickly identify business trends and improve operational efficiency and sustainability.

1.1 Product Positioning

> **"An integrated edge agent software that runs embedded within industrial-grade gateways and edge computers, with broad compatibility with mainstream industrial communication protocols, delivered through zero-code / low-code workflows."**

1.2 Target Markets

Industry	Typical Use Cases
Smart Power	Substation SCADA/telecontrol, meter reading, DTU/FTU, PV / storage monitoring
Smart Energy	BMS battery management, microgrid coordination, EMS energy management
Smart Manufacturing	PLC data acquisition, CNC connectivity, equipment OEE analytics, industrial cloud enablement
Building Automation	BACnet/Modbus acquisition, HVAC control, power monitoring
Smart Environmental	HJ212 emissions monitoring, pollution source tracking, water utility telemetry
Smart Water Utilities	Water meter acquisition, SL651 water resource telemetry, remote pump–station O&M
Smart Transportation	Tunnel monitoring, roadside equipment connectivity, EV charging station management
Oil & Gas	Modbus/OPC UA tank farm monitoring, oilfield data acquisition

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2. Key Features

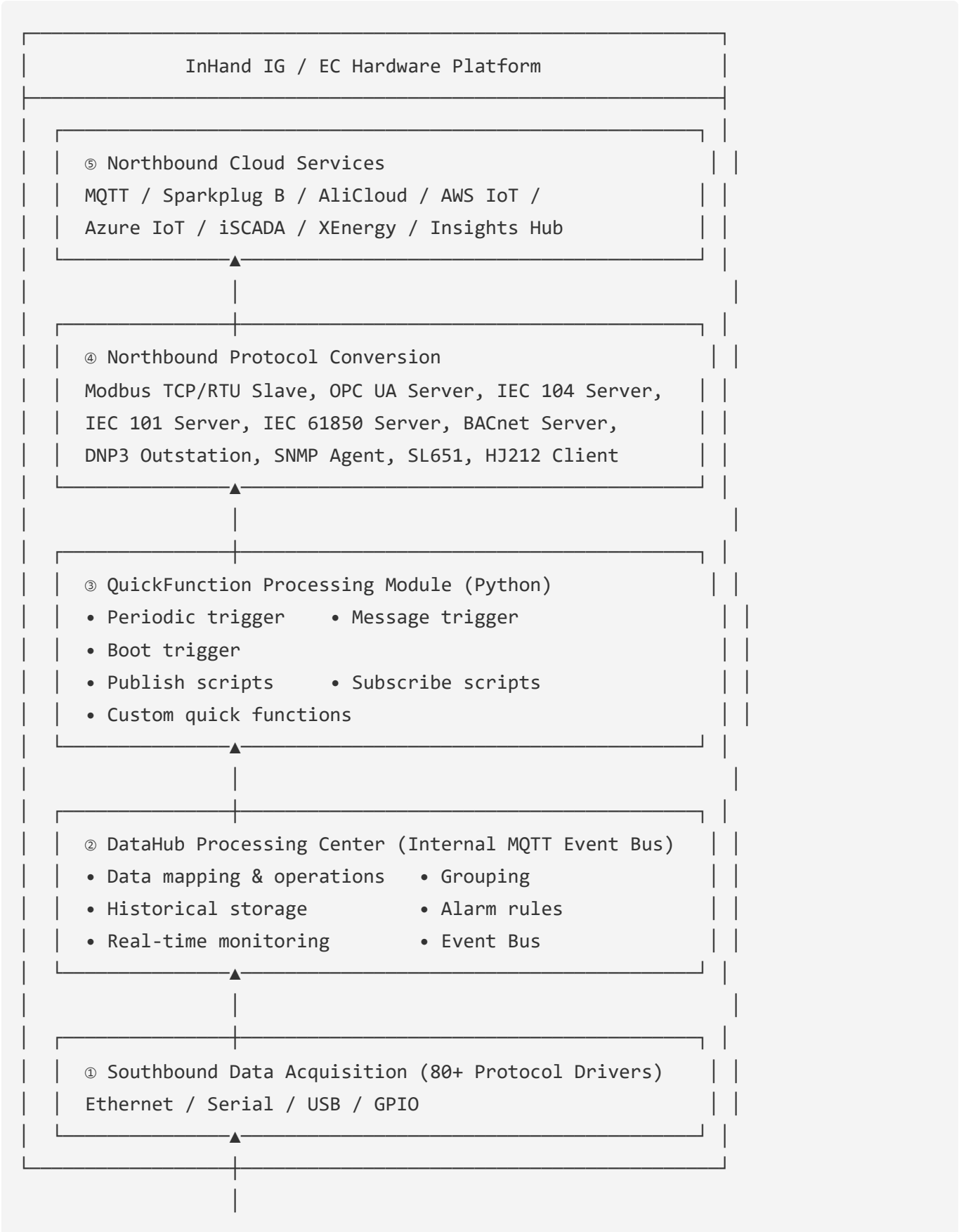
2.1 Ten Product Highlights

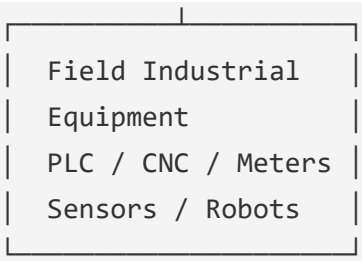
#	Feature	Description
1	**Rich Southbound Protocol Library**	Supports **80+ industrial protocols** covering PLCs, CNC, robots, SCADA, smart meters, and sensors

#	Feature	Description
2	**Extensive Northbound Protocol Conversion**	Supports **10+ northbound protocol servers** (Modbus / OPC UA / IEC 104 / BACnet / DNP3 / SNMP, etc.), eliminating upstream integration complexity
3	**Multi-Cloud Concurrent Publishing**	Simultaneously supports **AWS IoT, Azure IoT, AliCloud IoT, Siemens Insights Hub, InHand iSCADA / XEnergy** — one gateway publishes to multiple clouds in parallel
4	**Lightweight, Flexible Deployment**	Installs/uninstalls as an **APP/plug-in** on EC series products; low memory footprint; easy management
5	**Python Edge Intelligence**	Add Python scripts via **QuickFunction** for edge-side filtering, format conversion, analytics, and cloud-to-device control
6	**100ms High-Frequency Acquisition**	Per-controller and per-tag polling intervals; critical data acquired as fast as **100ms**
7	**Massive Tag Scalability**	Up to **190,000 tags per gateway** (EC942 hardware); concurrent multi-device polling
8	**Excel Bulk Configuration**	CSV/Excel bulk import/export of devices and tags; **device templates** for rapid deployment
9	**Local Historical Storage + Alarm Engine**	Built-in SQLite time-series database; **5-level alarms** (Notice/Warning/Minor/Major/Critical); offline caching and store-and-forward retransmission after recovery
10	**Real-Time Communication Diagnostics**	Southbound controllers support online viewing / downloading of communication messages for rapid on-site troubleshooting

3. System Architecture

DSA follows a **layered modular architecture** consisting of five main modules:





3.1 Module Descriptions

Module	Function
① Southbound Acquisition	80+ protocol drivers; supports Ethernet / RS485 / RS232; concurrent multi-device polling
② DataHub	Internal MQTT event bus for data mapping, operations, grouping, historical storage, and alarming
③ QuickFunction	Python scripting engine with periodic, message, and boot trigger modes
④ Northbound Protocol Conversion	10+ northbound server protocols with concurrent multi-protocol output
⑤ Northbound Cloud Services	Concurrent multi-cloud publishing including MQTT, Sparkplug B, AliCloud, AWS, Azure, etc.

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4. Southbound Protocol List (80+)

4.1 PLC Protocols (40+)

Brand	Supported Protocols
Rockwell / Allen-Bradley	EtherNet/IP (CompactLogix / Micro / MicroLogix / Unconnected), SLC Net

Brand	Supported Protocols
Siemens	ISO on TCP (S200 / S200 Smart / S1200 / S1500), PPI, PPI over TCP, Fetch/Write
Mitsubishi	FX Links (Serial/TCP), FX Serial over TCP, MC 1E/3E/3C, MC 3C over TCP, MC R, MITSUBISHI Programming Port
Omron	C-Mode (Serial/TCP), EtherNet/IP CompactLogix, FINS TCP/UDP, HostLink (Serial/TCP)
Beckhoff	ADS
Delta	Delta TCP, Delta Serial, Delta Serial ASCII, Delta Serial over TCP
FATEK	FATEK Programming Port (Serial/TCP)
Fuji	SPB (Serial/TCP), SPH Net
General Electric	SRTP
Inovance	INOVANCE Serial, INOVANCE TCP, INOVANCE Serial over TCP
KEBA	KEBA Socket
Keyence	KEYENCE MC 3E
Panasonic	Mewtocol (Serial/TCP), Panasonic MC 3E, Nano (Serial/TCP)
VIGOR	VIGOR Serial, VIGOR Serial over TCP
XINJE	XINJE TCP (Dedicated), XINJE TCP (Modbus), XINJE Serial, XINJE Serial over TCP
Yokogawa	Link TCP
Generic	Modbus RTU / ASCII / TCP / RTU over TCP / RTU Slave, OPC DA, OPC UA, OPC UA PubSub, Euromap-63 (Injection Molding), WebApi

4.2 Power Protocols (Chinese Power Standards)

- **DLT645-1997** (Chinese electric meter standard)
- **DLT645-2007** (Chinese electric meter standard)
- **DLT645-2007 over TCP**
- **DLT698** (next-generation Chinese meter standard)
- **DNP3 TCP / UDP / RTU** (North American mainstream)
- **IEC 101** (serial telemetry / telecontrol)
- **IEC 103** (NR Electric variant)
- **IEC 104** (Ethernet telemetry / telecontrol)
- **IEC 61850** (digitalized substation MMS messaging)

4.3 Building Automation Protocols

- **BACnet IP**
- **BACnet MS/TP**

4.4 CNC & Robots

- **ABB Robot**
- **EFORT Robot**
- **FANUC Robot**
- **FANUC CNC 0i**

4.5 Environmental & Water Utilities

- **HJ212-2005 Serial** (Chinese environmental protection standard)
- **HJ212-2017 Serial** (Chinese environmental protection standard)

4.6 Meters

- **CJT188-2004** (Chinese water meter standard)

4.7 Custom Protocols

- **Easy COM** — serial custom protocol acquisition
- **Easy TCP** — Ethernet custom protocol acquisition
- **Virtual Controller** — virtual controller for scripts / 3rd-party applications

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5. Northbound Protocol Conversion List (10+)

Expose acquired data via standard industrial protocols to upstream systems, such as SCADA, DCS, and HMI:

Northbound Protocol	Interface	Description
Modbus TCP Slave	TCP	Standard Modbus TCP slave for master polling
Modbus RTU Slave	RS232/485	Standard Modbus RTU slave
OPC UA Server	TCP	Standard OPC UA server for master polling
IEC 104 Server	TCP	Supports both TCP client and server modes
IEC 101 Server	RS232/485, TCP	Provides RS232/RS485/TCP transparent transmission
IEC 61850 Server	TCP	IEC 61850 MMS protocol with reporting functionality
BACnet IP Server	UDP	BACnet IP protocol for external acquisition
BACnet MS/TP Server	RS232/485	BACnet MS/TP protocol for external acquisition
DNP3 Outstation	TCP/UDP	Supports both TCP and UDP
SNMP Agent	UDP	SNMP protocol for external data reading
SL651–2014	TCP	Chinese Water Resources Ministry standard; data forwards to multiple hubs
HJ212 Client	TCP	Forwards data to multiple hubs using HJ212–2017

6. Northbound Cloud Service List

Cloud Platform	Type	Description
Standard MQTT	Generic	Docking with any standard MQTT broker
MQTT Sparkplug B	Standard	Industrial IoT data transfer specification built on MQTT 3.1.1
InHand iSCADA Cloud	InHand	Deep integration with InHand iSCADA cloud
InHand XEnergy	InHand	Deep integration with InHand XEnergy cloud
InHand XEnergy (IEC104)	InHand	Docking with XEnergy via IEC 104
AliCloud IoT	Alibaba	Docking with Alibaba Cloud IoT platform
AWS IoT	Amazon	Docking with AWS IoT Core
Azure IoT	Microsoft	Docking with Azure IoT Hub
Siemens MindSphere (Insights Hub)	Siemens	Docking with Siemens MindSphere / Insights Hub

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7. Edge Computing Capabilities (QuickFunction)

7.1 Three Trigger Modes

Trigger Mode	Use Case
Periodic Trigger	Executes at fixed intervals (seconds / minutes / hours); ideal for scheduled data processing, aggregation, and reporting

Trigger Mode	Use Case
Local Subscription Message Trigger	Subscribes to the internal MQTT event bus; triggers on message arrival — ideal for event-driven workflows
Boot Trigger	Executes once at DSA startup; ideal for initialization tasks

7.2 QuickFunction API Reference

API Category	Key Functions
Data Publish	Publish measurement points / alarms / custom data to cloud (publish QuickFunction)
Data Write	Update controller measurement point values (write)
Data Recall	Retrieve controller measurement point data (recall)
Global Parameters	Read global parameters, controller connection status, cloud connection status, global configuration
File I/O	Read / write files from within scripts
Time-Series DB	LwTSDB general API: insert / find / delete
Sparkplug B	Publish NBIRTH / NDATA; subscribe to NCMD

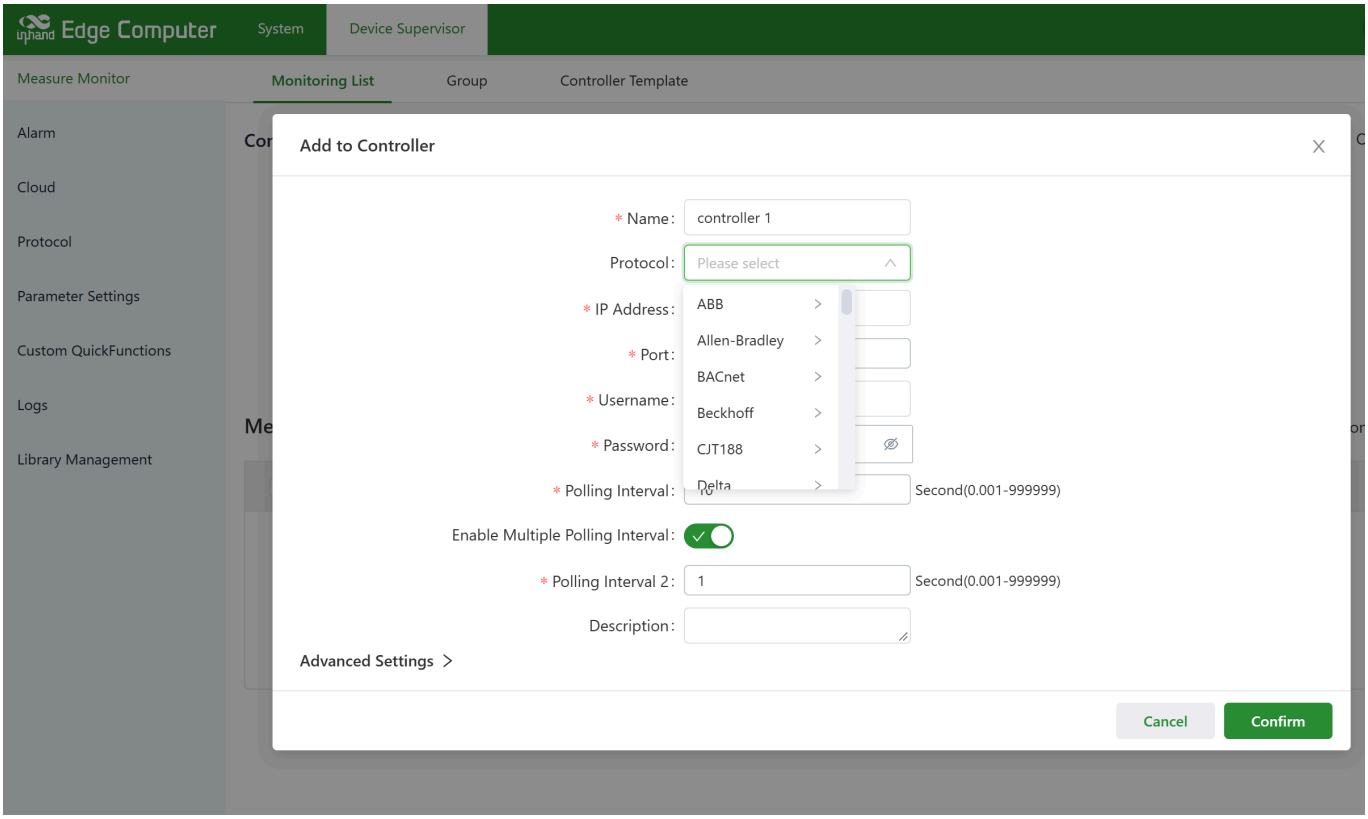
7.3 Internal MQTT Event Bus

Parameter	EC Series	IG Series
MQTT Broker Address	127.0.0.1	127.0.0.1
Port	(Docker network)	9009

Parameter	EC Series	IG Series
Southbound Read Topic	`ds2/eventbus/south/read/{driverServiceId}`	Same

8. Data Management Capabilities

8.1 Tag Configuration



Feature	Description
Controller Management	Add / edit / copy / delete / disable / enable controllers
Tag Management	Add / import / export / delete tags; Excel bulk configuration
Device Templates	Save common device configurations as templates; one-click deployment of identical devices

Feature	Description
Data Mapping & Operations	Data type conversion, byte-order adjustment, arithmetic operations, value mapping
Data Grouping	Group tags by business / scenario for easier reporting and querying
Real-Time Communication Messages	Online view / download of southbound controller messages for O&M troubleshooting

8.2 Data Type Support

Type	Byte Order
16 Bit Int	AB, BA (default AB)
32 Bit Int	ABCD, CDAB, DCBA, BADC (default ABCD)
32 Bit Float	ABCD, CDAB, DCBA, BADC (default ABCD)
64 Bit Int	ABCDEFGH, GHEFCDAB, HGEFDCBA, BADCFEHG (default ABCDEFGH)
64 Bit Float	ABCDEFGH, GHEFCDAB, HGEFDCBA, BADCFEHG (default ABCDEFGH)

8.3 Historical Data

- Built-in **SQLite3 time-series database**, storing historical data by group
- Per-tag storage frequency configuration
- Historical data CSV export

8.4 Alarm Engine

Parameter	Description
Alarm Levels	5 levels: Notice / Warning / Minor / Major / Critical
Comparison Operators	`=` , `!=` , `>` , `≥` , `<` , `≤`
Logical Operators	No logic / && (AND) / || (OR)
Alarm Labels	Group alarm items; MQTT reporting can select by label
Historical Alarms	Historical alarm records including trigger / restoration time and values
CSV Import / Export	Bulk alarm rule management

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9. Data Storage & Offline Cache & Retransmission

9.1 Storage Configuration

The parameters below configure offline caching and store-and-forward retransmission for cloud reporting.

Item	Parameter	Default
Log Level	DEBUG / INFO / WARNING / ERROR	INFO
Max Historical Alarms	Per-alarm maximum	2000
Offline Data Storage	USB / SD Card / Gateway Storage	Gateway Storage
Offline Data Max	Maximum records	10000
Offline Data Upload Period	ms	200

Item	Parameter	Default
Offline Data Report Timeout	ms	1000
Offline Data Report Strategy	Offline-priority / Realtime-priority	Offline-priority
Offline Data Retransmit Strategy	Continuous retry / Discard after limit	Continuous retry
Max Retry Count	Applied when "Discard after limit"	3
Communication Message Storage	USB / SD Card / Gateway Storage	Gateway Storage
Max Communication Messages		2000

9.2 Custom Parameters

Define **global parameters** for use in cloud service configuration and QuickFunction scripts:

- Built-in parameters: `SN` (device serial number), `MAC` (device MAC address)
- Add / import / export custom parameters
- Usage: `${parameter_name}` (as a wildcard placeholder)

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10. Performance Specifications

Measured DSA performance across InHand hardware platforms (single protocol, single controller, continuous tags):

10.1 IG502 (Entry-Level Industrial Gateway)

Protocol	Polling	Frame Gap	Tags	CPU
Modbus TCP	1s	0ms	6,000	100%

Protocol	Polling	Frame Gap	Tags	CPU
Modbus TCP	10s	10ms	40,000	100%
Modbus RTU	10s	10ms	40,000	100%
OPC UA	10s	–	38,000	85%

10.2 IG902 (Mainstream Industrial Gateway)

Protocol	Polling	Tags	CPU
Modbus TCP	10s	50,000	100%
Modbus RTU	10s	50,000	40–50%
OPC UA	10s	80,000	85%

10.3 IG532 (High-End Industrial Gateway)

Protocol	Polling	Tags	CPU
Modbus TCP	10s	48,000	100%
Modbus RTU	10s	48,000	65–75%
OPC UA	10s	45,000	70%

10.4 EC312 (Entry-Level Edge Computer)

Protocol	Polling	Tags	CPU
Modbus TCP	10s	45,000	65–85%
Modbus RTU	10s	45,000	40–50%
OPC UA	10s	42,400	100%

10.5 EC942 (Flagship Edge Computer)

Protocol	Polling	Tags	CPU
Modbus TCP	10s	**190,000**	60%
Modbus RTU	10s	**190,000**	24–30%
OPC UA	10s	80,000	30%

> **Performance Highlight**: A single EC942 supports ~190,000 tags — meets full-site data access needs of large sites; evaluate actual selection against project tag count, protocols, and refresh requirements.

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11. Hardware Compatibility

DSA is pre-installed on or available as an APP for the following InHand hardware platforms:

11.1 IG Series Industrial Gateways (DSA Pre-installed)

Model	Positioning	CPU	Memory	Typical Application
IG502	Entry 4G/5G industrial gateway	ARM Cortex-A8	512MB	Small equipment connectivity
IG902	Mainstream 4G/5G industrial gateway	ARM Cortex-A8	512MB-1GB	PLC acquisition, protocol conversion

11.2 EC Series Edge Computers (APP Installation)

Model	Positioning	CPU	Memory	Typical Application
EC312	Entry-level edge computer	ARM Cortex-A53	1GB	Building automation / water utilities

Model	Positioning	CPU	Memory	Typical Application
EC942	Flagship edge computer	ARM Cortex-A55	4GB/8GB	Large factories / substation edge

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12. System Requirements

Item	Requirement
Operating System	InHand custom Linux (kernel 5.x based)
CPU Architecture	ARM Cortex-A8 / A53 / A55
Minimum RAM	256MB (IG series) / 512MB (EC series)
Minimum Storage	512MB Flash
Network	Ethernet / 4G / 5G / Wi-Fi (per hardware)
Web Configuration Browser	Chrome 90+ / Edge 90+ / Firefox 88+

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13. Deployment & Operations

13.1 EC Series APP Installation

1. Log into the EC gateway web UI
2. Navigate to "Application Management" → "Upload Application"
3. Select the DSA package (`.tar.gz` format)
4. Click "Install" → resources are auto-configured
5. Configuration files auto-load upon startup

13.2 IG Series Built-in Operation

1. IG series ships with DSA pre-installed
2. Log into the IG gateway web UI

3. Navigate to "Edge Computing" → "DeviceSupervisor"
4. Directly configure controllers, tags, and cloud services

13.3 Configuration Updates

- Online update of DSA configuration files (JSON format)
- Web UI version upgrade
- Bulk deployment via InHand iSCADA cloud platform

13.4 Log Diagnostics

- Online view / download of runtime logs
- Configurable log levels: DEBUG / INFO / WARNING / ERROR
- Critical log power-loss preservation

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14. Licensing & Ordering

14.1 Licensing Models

License Model	Description
Hardware Embedded	IG series ships with full DSA functionality; no separate license required
EC Series APP	EC series can install DSA as an APP with corresponding license
Enterprise Subscription	Enterprise annual subscription available for large-scale deployments, including technical support and cloud management

14.2 Services & Support

Service	Content
Technical Documentation	User manuals, API references, protocol integration guides

Service	Content
Online Knowledge Base	help.inhand.com — 24/7 access
Technical Support	Email / phone / remote support
Software Upgrades	Free major version upgrades (within license validity)
Protocol Extension	Custom new protocols available upon customer request

14.3 Ordering Information

Please contact InHand Networks sales for pricing and ordering:

- **China Region**: sales@inhand.com.cn
- **Overseas**: sales@inhand.com
- **Website**: www.inhand.com

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15. About InHand Networks

Founded in 2001, InHand Networks is a leading global provider of Industrial IoT solutions, dedicated to driving digital transformation across industries. Our business spans **Enterprise Networking, Industrial & Building IoT, Digital Energy, Smart Commerce, and Intelligent Transportation**, with products and services deployed in 60+ countries including China, the United States, France, Germany, the United Kingdom, and Italy.

Headquarters: Beijing, China **Overseas Offices**: USA, Singapore, Germany, Canada **Website**: www.inhand.com

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Appendix A: Differentiation Quick Reference vs. Competitors

Capability	DSA	Competitor with same capability
Chinese power protocols (DLT645/IEC61850)	✅ Native	Kepware ✅
Chinese environmental protocols (HJ212)	✅ **Industry Exclusive**	None
100ms high-frequency acquisition	✅	Kepware / Litmus ✅
Python scripting	✅	Robustel / EdgeLink ✅
Excel bulk configuration	✅	EdgeLink ✅
Hardware-embedded, zero integration	✅	Robustel / Advantech ✅
5-level alarm engine	✅	Advantech ✅
Multi-cloud concurrent publishing	✅	Litmus ✅
Tag scale (EC942)	**190,000**	Hardware-dependent

Appendix B: Revision History

Version	Date	Changes
V1.0	2026-09	Initial release based on DSA V3.1.10

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